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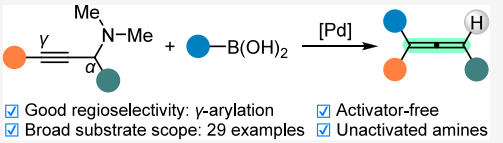
Assembly of Diverse Allenes via Activator-Free Palladium-Catalyzed Regioselective γ-Arylation of Propargylamines with Boronic Acids

Yan Sun,[§] Ya Pan,[§] Tao Zhao,* Junwei Li, Aiguo Gu, Junhai Huang, and Huangdi Feng*

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ABSTRACT: Controlling the reaction selectivity to achieve a precision synthesis is a constant concern for chemists. Here, we report a palladium-catalyzed deaminative coupling of propargylamines with arylboronic acids to generate allene skeletons. Importantly, this approach allows the regioselective γ-arylation of unactivated propargyl tertiary amines to access various allenes in the absence of amino-activating reagents. We present a wide range of propargylamines and boronic acids and demonstrate the synthetic application of the target products to construct valuable compounds.



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INTRODUCTION

The amine family, a wide range of building blocks, has been extensively used as synthetic handles in synthetic chemistry and medicinal chemistry.[1] Although amines offer immense synthetic leverage, the direct deaminative transformations of tertiary amines remain underutilized because the amino group, as a low heterolytic nucleofugal leaving group, has a high homolytic C−N bond dissociation energy and stronger characteristic basicity.[2] Accordingly, much effort has been spent toward the deaminative functionalization of unactivated tertiary amines. The classical method for C−N functionalization involves the conversion of tertiary amines to ammonium salts.[3] This preactivation strategy requires at least two distinct transformations to achieve C−N functionalization. In addition, in situ activation strategies such as the catalytic oxidation of tertiary amines to form iminium intermediates[4] or the use of activating agents[5] to induce deaminative functionalization have been demonstrated. However, each of these previous reports required the use of a large excess of oxidant or activator, producing chemical waste that does not meet the demand for green chemistry. Currently, many research groups have focused on a variety of transition metal-catalyzed[6] and metal-free[7] strategies but are generally limited to aryl and benzylic tertiary amines in cross-coupling reactions. To date, a few examples of direct deaminative transformations of tertiary amines are stark reminders of the shortcomings in this area of synthetic chemistry. Therefore, developing a novel, efficient, and atom-economic protocol for the C−N functionalization of tertiary amines is highly important.

On the other hand, allenes have proven to be efficient and versatile synthons in various cycloaddition reactions, providing access to valuable molecules with medicinal, agrochemical, and material significance.[8] They are also important scaffolds widely found in natural products, bioactive compounds, and marketed drugs.[9] In this context, a great deal of efforts have been devoted to the development of efficient approaches for the construction of allene moieties.[10] In this regard, the transition metal-catalyzed deaminative reaction has emerged as one of the most attractive methods for the synthesis of allenes.[11] Among these deaminative strategies, the C−N cleavage of propargyl amines has been widely investigated, and various metal catalysts such as Zn, Cu, In, Cd, Ag, Au, and Pd have been employed (Scheme 1A).[12] However, these previous reports mainly rely upon the use of specific substrates and limit the structural diversity of allenes. The catalytic cross-coupling of electrophilic propargyl derivatives with various nucleophiles has offered a promising approach for designing structural diversity and complexity.[13] However, the reported deaminative strategies to access allenes are limited to the utilization of propargyl ammonium salts and organometallic reagents through a process of γ-alkylation[13b] (Scheme 1B). To address this shortfall, difluorocarbene-induced and alkyne-mediated deaminative couplings of unactivated propargylamines and boronic acids have been successfully developed by Song's group[14a] and our group,[14b] respectively, yielding propargylic substitution products rather than allenes (Scheme 1C). The exploitation of catalytic deaminative coupling methods to access allenes using simple propargylamines and nucleophiles remains underdeveloped, with the main challenge being selective control to avoid the propargylic substitution reaction. As part of our ongoing investigation into propargylamine functionalizations, we report herein a novel palladium-catalyzed deaminative coupling of propargyl tertiary amines and arylboronic acids to construct diverse allenes via a regioselective γ-arylation process (Scheme 1D).

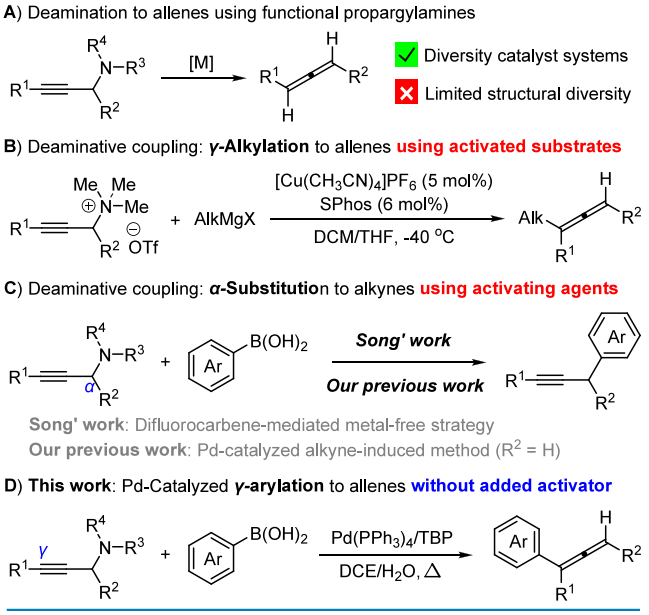
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Scheme 1. Deaminative Strategies for Propargylamine Transformations[a]



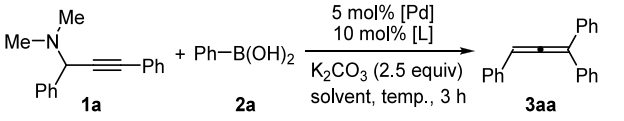
[a](A–C) Previous reports; (D) our work.

RESULTS AND DISCUSSION

We initially discovered that the introduction of a substituent at the α-position of the nitrogen atom was highly effective in promoting the deaminative γ-arylation of propargylamines. Thus, we started our initial screen by using *N*,*N*-dimethyl-1,3-diphenylprop-2-yn-1-amine **1a** and phenylboronic acid **2a** as model substrates in the presence of $PdCl_2(PPh_3)_2$ (5 mol %), tributylphosphine (TBP; 10 mol %), and K_2CO_3 (2.5 equiv) in 1,2-dichloroethane (DCE; 2 mL) at 120 °C for 3 h to afford allene product **3a** in a 39% yield (Table 1, entry 1). Subsequently, various commercially available Pd catalysts including $Pd_2(dba)_3$, $Pd(PPh_3)_4$, and $Pd(OAc)_2$ were tested for this reaction (entries 2–4). To our delight, the yield of **3a** was increased to 49% when $Pd(PPh_3)_4$ was used. Different ligands were also examined in this coupling reaction under the catalysis of $Pd(PPh_3)_4$ (entries 5–7), but none of them could improve the yield. In general, the trialkylphosphines (PEt_3 and PCy_3) served as ligands and gave higher yields than those with an aromatic group such as 2-dicyclohexylphosphino-2′,4′,6′-triisopropylbiphenyl (XPhos). Taking advantage of the addition of water to enhance the reaction,[14b,15] we then screened a number of solvents (fluorobenzene, trifluoromethylbenzene, toluene, and 1,4-dioxane) with the addition of water (entries 8–12). It is obvious that the combination of DCE and water is the most efficient for this deaminative transformation. We were pleased to find that the yield of allene **3a** was further improved to 74% when the amount of phenylboronic acid **2a** was increased to 2.5 equiv (entry 13). Lowering the reaction temperature to 100 °C resulted in a significant decrease in the yield of **3a** (entry 14, 56%). In addition, other experimental parameters such as the type and loading of the base, the amount of water added, and the reaction time were also evaluated, but all led to a negative result (for details, see the Supporting Information, Table S1).

With the optimized conditions in hand, we next investigated the generality of the reaction substrates (Scheme 2). First, we

Table 1. Reaction Optimization for Coupling of 1a with 2a[a]



entry	catalyst	ligand	solvent	temp (°C)	yield[b] (%)
1	$PdCl_2(PPh_3)_2$	TBP	DCE	120	39
2	$Pd_2(dba)_3$	TBP	DCE	120	trace
3	$Pd(PPh_3)_4$	TBP	DCE	120	49
4	$Pd(OAc)_2$	TBP	DCE	120	21
5	$Pd(PPh_3)_4$	PEt_3	DCE	120	40
6	$Pd(PPh_3)_4$	PCy_3	DCE	120	39
7	$Pd(PPh_3)_4$	XPhos	DCE	120	34
8[c]	$Pd(PPh_3)_4$	TBP	DCE/H_2O	120	69
9[c]	$Pd(PPh_3)_4$	TBP	PhF/H_2O	120	56
10[c]	$Pd(PPh_3)_4$	TBP	$PhCF_3$/H_2O	120	60
11[c]	$Pd(PPh_3)_4$	TBP	PhMe/H_2O	120	37
12[c]	$Pd(PPh_3)_4$	TBP	dioxane/H_2O	120	n.d.
13[c,d]	$Pd(PPh_3)_4$	TBP	DCE/H_2O	120	74
14[c,d]	$Pd(PPh_3)_4$	TBP	DCE/H_2O	100	56

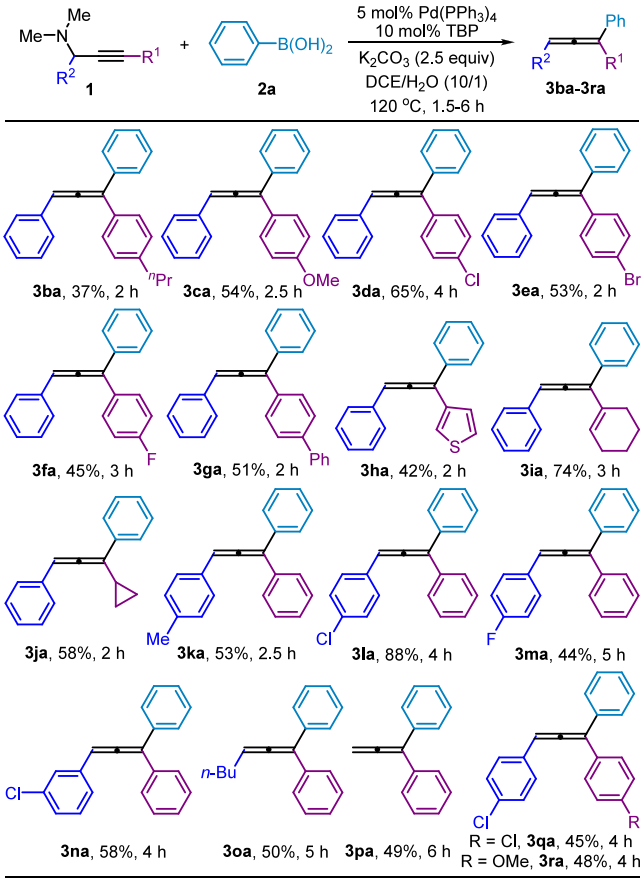
[a]Reaction conditions: **1a** (0.2 mmol), **2a** (2 equiv), Pd catalyst (5 mol %), ligand (10 mol %), K_2CO_3 (0.5 mmol), solvent (2 mL) at 100–120 °C for 3 h under atmospheric conditions. n.d., not detected. [b]Isolated yield based on **1a**. [c]Solvent/H_2O (v/v) = 10/1. [d]Using 2.5 equiv of **2a**. XPhos = 2-dicyclohexylphosphino-2′,4′,6′-triisopropylbiphenyl; TBP=tri-*n*-butylphosphine; DCE = 1,2-dichloroethane.

investigated the extent to which propargylamines **1** containing different R^1 and R^2 substituent groups would react with phenylboronic acid **2a**. Propargylamines **1** containing both aromatic and aliphatic R^1 groups were smoothly converted to a small library of trisubstituted allenes **3ba–3ja** in moderate to good yields. For example, various *para*-substituted aromatic groups, including *n*-Pr, MeO, Cl, Br, F, and Ph substitutions as well as the thiophene group, all reacted smoothly. Analogously, aliphatic groups such as cyclohexenyl and cyclopropyl substitutions were also tolerated, providing the target products **3ia** and **3ja** in 74 and 58% yields, respectively. The generality of R^2 groups on propargylamines **1** was then evaluated. In general, different R^2-substituted propargylamines **1k–1p** reacted well in this deaminative functionalization process. When the R^2 substitution was the substituted phenyl groups, 44–88% yields of the desired products **3ka–3na** were obtained. Other R^2 groups including the *n*-butyl group and hydrogen could be converted to the desired allenes **3oa** and **3pa** in moderate yields. In addition, the modification of both R^1 and R^2 groups was found to be efficient and afforded the corresponding products **3qa** and **3ra** in reasonable yields.

Next, various boronic acids **2** were tested for their effect on deaminative coupling with **1a** (Scheme 3). In this transformation, all aromatic boronic acids afforded the desired products **3ab–3al** in moderate to good yields. For example, both electron-rich (Me, Et, *t*-Bu, and OMe groups) and electron-deficient (Cl and F groups) aryl boronic acids afforded the corresponding allenes **3ab–3ag** in 48–62% yields. Electron-neutral aromatic boronic acids including vinyl- and phenyl-substituted phenylboronic acids as well as naphthyl boronic acids were compatible with the reaction systems, providing the target products **3ah–3ak** in 36–87% yields. In particular, heterocyclic boronic acid was found to be a suitable substrate for deaminative coupling to form product

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Scheme 2. Substrate Scope of Propargylamines[a]



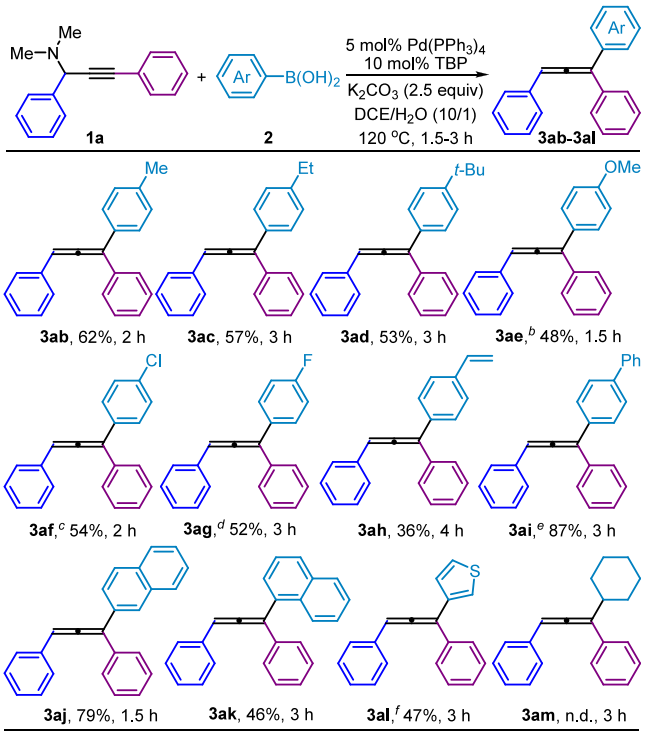
[a]Reaction conditions: **1** (0.2 mmol), **2a** (0.5 mmol), $Pd(PPh_3)_4$ (5 mol %), TBP (10 mol %), K_2CO_3 (0.5 mmol), DCE/H_2O (v/v = 10/1) at 120 °C for 1.5−6 h under atmospheric conditions. Isolated yield based on **1**.

3al. However, the deaminative coupling of cyclohexylboronic acid with **1a** did not yield the target product **3am**. In addition, we also tried other boronic acids, such as (*E*)-styrylboronic acid, (4-formylphenyl)boronic acid, *n*-pentylboronic acid, and pyridine-4-boronic acid. However, all of them did not generate the target products (for details, see the Supporting Information, Table S2).

Finally, the reaction on a 2 mmol scale was carried out without any further optimization but afforded a somewhat diminished yield for product **3aa** (Scheme 4a). We also adjusted the amount of catalyst and the amount of solvent, but unfortunately, the yield did not improve significantly (for details, see the Supporting Information, Table S3). Furthermore, the synthetic utility of the obtained product was demonstrated by the following transformations (Scheme 4b).[16] Utilizing a previous method[16a] and starting from **3aa**, the indene compound **4** was efficiently constructed in a yield of 41% by TfOH-promoted cyclization. The treatment of **3aa** using the $(PPh_3)AuCl/AgBF_4$ catalyst system gave the oxidation product **5** in a yield of 69%.[16b]

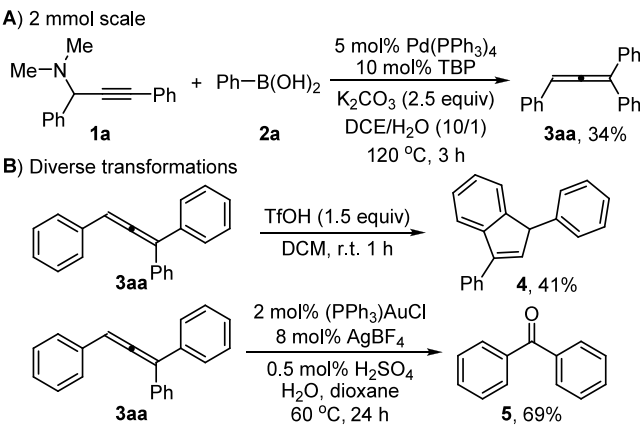
On the basis of the previous literature[17] and our studies, a plausible mechanism is proposed as shown in Scheme 5. First, the oxidative addition of Pd(0) species and propargylamine **1** forms the allenylpalladium complex **A** through $\sigma-\pi-\sigma$ rearrangement. Subsequently, the allenylic palladium species **A** is converted to the Pd hydroxide intermediate **B** by reaction

Scheme 3. Substrate Scope of Boronic Acids[a]



[a]Reaction conditions: **1a** (0.2 mmol), **2** (0.5 mmol), $Pd(PPh_3)_4$ (5 mol %), TBP (10 mol %), K_2CO_3 (0.5 mmol), DCE/H_2O (v/v = 10/1) at 120 °C for 1.5−3 h under atmospheric conditions. Isolated yield based on **1a**. [b]**3ae** = **3ca**. [c]**3af** = **3da**. [d]**3ag** = **3fa**. [e]**3ai** = **3ga**. [f]**3al** = **3ha**.

Scheme 4. Scalability and Product Transformations[a]



[a](A) Millimole-scale synthesis; (B) transformations of product **3aa**.

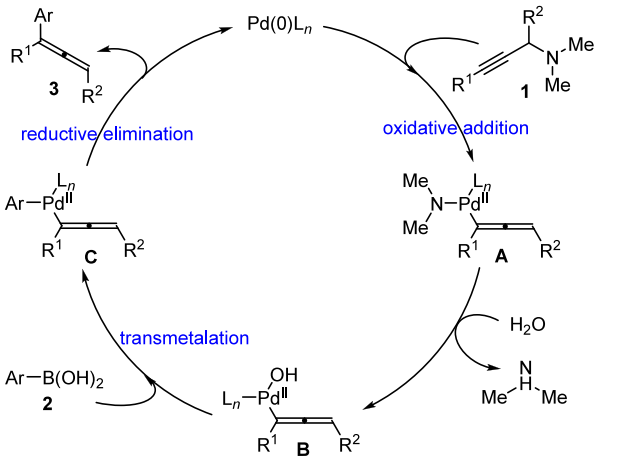
with water under basic conditions,[18] followed by the transmetalation with boronic acid **2** to deliver the corresponding allenylic intermediate **C**. Finally, the allene product **3** is generated via reductive elimination of intermediate **C** and the catalytically active Pd(0) species is regenerated.

CONCLUSIONS

In summary, a straightforward synthetic protocol has been developed for the preparation of trisubstituted allenes via the palladium-catalyzed deaminative coupling of unactivated propargylamines and aryboronic acids. Importantly, we demonstrate for the first time that this catalytic deaminative

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Scheme 5. Proposed Mechanism



coupling reaction can be achieved in the absence of any external activator. Compared to the traditional catalytic deaminative coupling via in situ amino activation, the activator-free method offers some sustainable features in terms of operational simplicity, good functional group tolerance with high yield and regioselectivity, and the absence of toxic byproducts. Efforts to explore new synthetic applications for activator-free catalytic deaminative transformation are currently underway in our laboratory.

EXPERIMENTAL SECTION

General Information. Unless otherwise noted, all commercial reagents, such as boronic acids, catalyst, ligands, and solvent, were used directly as purchased. All reactions were carried out under air. Thin-layer chromatography (TLC) was performed, and visualization of the compounds was accomplished with ultraviolet light (254 nm) or iodine. Products were purified by flash chromatography on silica gel (100−200 mesh). The solvent was removed under reduced pressure, and the residue was purified by column chromatography on silica gel and eluted with petroleum/ethyl acetate to afford the desired product.^{1}H nuclear magnetic resonance (NMR), ^{13}C NMR, and ^{19}F spectra were recorded in $CDCl_3$ operating at 400, 101, and 376 MHz, respectively. Proton chemical shifts are reported relative to the residual proton signals of deuterated solvent $CDCl_3$ (7.29 ppm). Carbon chemical shifts were internally referenced to the deuterated solvent signals in $CDCl_3$ (77.10 ppm). Chemical shifts are reported in δ (parts per million) values. Coupling constants J are reported in Hz. Proton coupling patterns were described as singlet (s), doublet (d), triplet (t), quartet (q), and multiple (m). High-resolution mass spectrometry (HRMS) was recorded on a Q-TOF (AB SCIEX X500R with an electrospray ionization (ESI) source and Agilent 7250 with an electron ionization (EI) source), which combines quadrupole precursor ion selection and a high-resolution accurate-mass time-of-flight mass analyzer to deliver mass accuracy.

General Procedure for the Synthesis of the Desired Product 3. $Pd(PPh_3)_4$ (11.6 mg, 0.01 mmol, 5 mol %), TBP (4.0 mg, 0.02 mol, 10 mol %), K_2CO_3 (0.5 mmol, 2.5 equiv), propargylamines **1** (0.2 mmol), and arylboronic acids **2** (0.5 mmol, 2.5 equiv) were added to a 5 mL Schlenk reaction tube, followed by the addition of DCE (2 mL) and water (0.2 mL). The mixture was stirred in an oil bath and heated to 120 °C (oil bath temperature) for 1.5−6 h in reflux. After the reaction had reached completion (as determined by TLC), the reaction mixture was cooled to room temperature and extracted with DCM. It was then filtered, and the solvent was removed in a rotary evaporator. The residue was purified by flash chromatography (petroleum ether) to provide desired products **3aa**.

2 mmol of the Scaled Reaction. $Pd(PPh_3)_4$ (115.6 mg, 0.1 mmol, 5 mol %), TBP (40.5 mg, 0.2 mol, 10 mol %), K_2CO_3 (5.0 mmol, 2.5 equiv), propargylamines **1** (2.0 mmol), and arylboronic acids **2** (5 mol, 2.5 equiv) were added to a 50 mL single-necked flask, followed by the addition of DCE (10 mL) and water (1.0 mL). The mixture was stirred in an oil bath preheated to 120 °C for 3 h (monitored by TLC). After the reaction had reached completion (as determined by TLC), the reaction mixture was cooled to room temperature and extracted with DCM. It was then filtered, and the solvent was removed in a rotary evaporator. The residue was purified by flash chromatography (petroleum ether) to provide the desired products **3** (34%, 184.5 mg).

General Procedure for the Synthesis of the Desired Product 4.[19] To a suspension of trifluoromethylated allene **3aa** (53.7 mg, 0.2 mmol) in CH_2Cl_2 (2 mL) was added TfOH (18 μL, 0.15 mmol) at room temperature. The resulting mixture was stirred at room temperature for 1 h. Then, the mixture was poured into ice water (20 mL) and extracted with CH_2Cl_2. The combined organic layers were washed with saturated $NaHCO_3$ (aq.), dried over $MgSO_4$, and concentrated in vacuo. The residue was purified by column chromatography using n-hexane as an eluent to afford 1,3-diphenyl-1H-indene **4** (22.0 mg, 41% yield).

General Procedure for the Synthesis of the Desired Product 5.[16b] Sulfuric acid (0.5 mol %) was added to a mixture of **3aa** (0.2 mmol), water (4 mmol), $(PPh_3)AuCl$ (2 mol %), $AgBF_4$ (8 mol %), and dioxane (1 mL). The mixture was stirred at 60 °C for 24 h, and the reaction was quenched with a saturated solution of $NaHCO_3$ and extracted with ethyl acetate (3 × 10 mL). The combined organic layers were washed with brine, dried over Na_2SO_4, and concentrated in vacuo. The residue was purified by flash chromatography to give desired products **5** (25.1 mg, 69%).

*Propa-1,2-diene-1,1,3-triyltribenzene (**3aa**).*[20] Purification by silica gel column chromatography (petroleum ether (PE)) afforded **3aa** (53.1 mg, 74% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.47−7.38 (m, 6H), 7.38−7.26 (m, 8H), 7.25−7.19 (m, 1H), 6.71 (s, 1H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 208.4, 135.9, 133.0, 132.5, 129.0, 128.6, 128.5, 128.2, 127.8, 114.2, 96.9.

*(1-(4-Propylphenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3ba**).* Purification by silica gel column chromatography (PE) afforded **3ba** (23.0 mg, 37% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.45−7.39 (m, 4H), 7.38−7.27 (m, 7H), 7.24−7.17 (m, 1H), 7.15 (d, J = 8.2 Hz, 2H), 6.69 (s, 1H), 2.62−2.55 (m, 2H), 1.68−1.60 (m, 2H), 0.95 (t, J = 7.4 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 208.2, 142.3, 136.4, 134.1, 133.4, 131.5, 128.8, 128.7, 128.5, 128.5, 128.3, 127.6, 127.3, 127.0, 113.7, 97.6, 37.8, 24.6, 13.9. HRMS (EI) m/z: [M$^+$] calcd for $C_{24}H_{22}$, 310.1717; found, 310.1715.

*(1-(4-Methoxyphenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3ca**).*[20] Purification by silica gel column chromatography (PE) afforded **3ca** (32.2 mg, 54% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.46−7.38 (m, 4H), 7.38−7.27 (m, 7H), 7.25−7.19 (m, 1H), 6.91−6.86 (m, 2H), 6.68 (s, 1H), 3.81 (s, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 208.0, 159.2, 136.5, 134.2, 129.6, 128.8, 128.5, 128.5, 128.3, 127.6, 127.3, 127.0, 114.0, 97.6, 55.4.

*(1-(4-Chlorophenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3da**).* Purification by silica gel column chromatography (PE) afforded **3da** (39.4 mg, 65% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.43−7.38 (m, 4H), 7.37−7.27 (m, 10H), 6.66 (s, 1H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 208.3, 135.8, 134.7, 133.6, 133.4, 129.7, 128.9, 128.7, 128.6, 128.4, 127.8, 127.6, 127.0, 112.9, 98.1. HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{21}H_{16}Cl$, 303.0936; found, 303.0930.

*(1-(4-Bromophenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3ea**).*[21] Purification by silica gel column chromatography (PE) afforded **3ea** (36.8 mg, 53% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.47−7.43 (m, 2H), 7.41−7.26 (m, 11H), 7.25−7.20 (m, 1H), 6.70 (s, 1H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 208.3, 135.7, 135.3, 133.5, 131.7, 130.1, 128.9, 128.6, 128.4, 127.9, 127.6, 127.0, 121.6, 113.0, 98.1.

*(1-(4-Fluorophenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3fa**).* Purification by silica gel column chromatography (PE) afforded **3fa** (25.8 mg, 45% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.42−7.27 (m, 11H), 7.25−7.20 (m, 1H), 7.06−6.99 (m, 2H), 6.70

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(s, 1H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.1, 163.6, 161.1, 136.1, 133.8, 132.2 (d, J = 3.2 Hz), 130.1 (d, J = 8.1 Hz), 128.9, 128.6, 128.4, 127.8, 127.5, 127.0, 115.6, 115.4, 112.9, 97.9. ^{19}F NMR (376 MHz, $CDCl_3$) δ −113.17. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{21}H_{16}F$, 287.1231; found, 287.1235.

*4-(1,3-Diphenylpropa-1,2-dien-1-yl)-1,1′-biphenyl (**3ga**).*[20] Purification by silica gel column chromatography (PE) afforded **3ga** (35.1 mg, 51% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.61−7.55 (m, 3H), 7.53−7.40 (m, 8H), 7.39−7.28 (m, 6H), 7.25−7.20 (m, 2H), 6.73 (s, 1H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.5, 140.8, 140.5, 136.2, 135.1, 133.9, 128.9, 128.8, 128.57, 127.7, 127.5, 127.4, 127.3, 127.1, 113.5, 97.9.

*3-(1,3-Diphenylpropa-1,2-dien-1-yl)thiophene (**3ha**).* Purification by silica gel column chromatography (PE) afforded **3ha** (23.0 mg, 42% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.51−7.47 (m, 2H), 7.42−7.30 (m, 8H), 7.25−7.19 (m, 2H), 7.20−7.14 (m, 2H)., 6.68 (s, 1H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.3, 136.7, 136.3, 133.9, 128.8, 128.8, 128.6, 128.2, 128.0, 127.8, 127.4, 127.1, 125.7, 122.8, 109.3, 97.6. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{19}H_{15}S$, 275.0889; found, 275.0886.

*(1-(Cyclohex-1-en-1-yl)propa-1,2-diene-1,3-diyl)dibenzene (**3ia**).* Purification by silica gel column chromatography (PE) afforded **3ia** (40.3 mg, 74% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.39−7.27 (m, 8H), 7.27−7.22 (m, 1H), 7.21−7.16 (m, 1H), 6.50 (s, 1H), 5.72−5.64 (m, 1H), 2.29−2.08 (m, 4H), 1.74−1.60 (m, 4H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 206.8, 136.5, 134.9, 132.9, 129.1, 128.7, 128.2, 127.6, 127.2, 127.0, 126.8, 115.6, 97.0, 27.6, 26.0, 22.9, 22.3. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{21}H_{21}$, 273.1638; found, 273.1640.

*(1-Cyclopropylpropa-1,2-diene-1,3-diyl)dibenzene (**3ja**).*[22] Purification by silica gel column chromatography (PE) afforded **3ja** (26.9 mg, 58% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.64−7.59 (m, 2H), 7.43−7.36 (m, 2H), 7.36−7.30 (m, 3H), 7.30−7.26 (m, 4H), 6.54 (d, J = 2.6 Hz, 1H), 1.74−1.64 (m, 1H), 0.90 (d, J = 8.0 Hz, 2H), 0.67−0.55 (m, 2H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 205.8, 136.5, 134.31, 128.8, 128.5, 127.2, 127.2, 126.76, 126.4, 113.3, 99.0, 11.1, 7.4, 6.9.

*(3-(p-Tolyl)propa-1,2-diene-1,1-diyl)dibenzene (**3ka**).*[23] Purification by silica gel column chromatography (PE) afforded **3ka** (29.9 mg, 53% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.44−7.39 (m, 4H), 7.37−7.25 (m, 8H), 7.13 (d, J = 7.9 Hz, 2H), 6.68 (s, 1H), 2.33 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.2, 137.3, 136.4, 130.9, 129.6, 128.5, 127.5, 127.4, 127.3, 126.9, 113.6, 97.6, 21.3.

*(3-(4-Chlorophenyl)propa-1,2-diene-1,1-diyl)dibenzene (**3la**).*[20] Purification by silica gel column chromatography (PE) afforded **3la** (53.3 mg, 88% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.42−7.38 (m, 4H), 7.38−7.34 (m, 3H), 7.34−7.26 (m, 7H), 6.66 (s, 1H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.4, 135.9, 133.0, 132.5, 129.0, 128.6, 128.5, 128.2, 127.8, 114.2, 96.9.

*(3-(4-Fluorophenyl)propa-1,2-diene-1,1-diyl)dibenzene (**3ma**).*[20] Purification by silica gel column chromatography (PE) afforded **3ma** (34.0 mg, 44% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.43−7.28 (m, 11H), 7.26−7.20 (m, 1H), 7.03 (t, J = 8.8 Hz, 2H), 6.70 (s, 1H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.0, 163.4, 161.0, 136.1, 129.9 (d, J = 3.2 Hz), 128.5 (d, J = 8.8 Hz), 128.4, 127.7, 115.9, 115.7, 114.0, 96.8. ^{19}F NMR (376 MHz, $CDCl_3$) δ −114.62.

*(3-(3-Chlorophenyl)propa-1,2-diene-1,1-diyl)dibenzene (**3na**).* Purification by silica gel column chromatography (PE) afforded **3na** (35.1 mg, 58% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.43−7.38 (m, 4H), 7.38−7.27 (m, 8H), 7.27−7.22 (m, 2H), 7.22−7.17 (m, 1H), 6.65 (s, 1H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.5, 136.0, 135.8, 134.8, 130.3, 130.0, 128.6, 128.5, 127.8, 127.4, 126.8, 125.1, 114.3, 96.8. HRMS m/z (ESI-TOF): calculated for $C_{21}H_{16}Cl$ ($[M + H]^+$), 303.0936; found, 303.0930.

*Hepta-1,2-diene-1,1-diyldibenzene (**3oa**).*[20] Purification by silica gel column chromatography (PE) afforded **3oa** (24.8 mg, 50% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.37−7.29 (m, 8H), 7.27−7.22 (m, 2H), 5.69 (t, J = 6.7 Hz, 1H), 2.19 (q, J = 7.1 Hz, 2H), 1.55−1.45 (m, 2H), 1.43−1.33 (m, 2H), 0.89 (t, J = 7.4 Hz, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 205.5, 137.4, 137.4, 128.4, 128.3, 128.2, 128.1, 127.0, 98.0, 94.5, 31.4, 28.8, 22.4, 14.0.

*Propa-1,2-diene-1,1-diyldibenzene (**3pa**).*[23] Purification by silica gel column chromatography (PE) afforded **3pa** (18.8 mg, 49% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.40−7.23 (m, 10H), 5.27 (s, 2H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 207.8, 136.2, 133.6, 128.7, 128.4, 128.4, 127.3, 127.0, 98.4.

*4,4′-(1-Phenylpropa-1,2-diene-1,3-diyl)bis(chlorobenzene) (**3qa**).* Purification by silica gel column chromatography (PE) afforded **3qa** (30.4 mg, 45% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.40−7.34 (m, 5H), 7.33−7.29 (m, 8H), 7.26 (s, 1H), 6.68 (s, 1H), 1.32−1.27 (m, 1H), 1.25 (s, 4H), 0.91−0.80 (m, 2H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.3, 135.5, 133.6, 133.2, 132.1, 129.7, 129.1, 128.8, 128.7, 128.4, 128.2, 128.0, 113.3, 97.2, 29.7. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{21}H_{15}Cl_2$, 337.0546; found, 337.0548.

*1-Chloro-4-(3-(4-methoxyphenyl)-3-phenylpropa-1,2-dien-1-yl)-benzene (**3ra**).* Purification by silica gel column chromatography (PE) afforded **3ra** (32.0 mg, 48% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.43−7.38 (m, 2H), 7.38−7.25 (m, 12H), 6.64 (s, 1H), 3.82 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.1, 159.3, 136.2, 133.0, 132.9, 132.7, 129.6, 129.3, 129.0, 128.6, 128.6, 128.5, 128.1, 128.0, 127.7, 114.0, 96.7, 55.4. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{22}H_{18}ClO$, 333.1041; found, 333.1044.

*(1-(p-Tolyl)propa-1,2-diene-1,3-diyl)dibenzene (**3ab**).*[24] Purification by silica gel column chromatography (PE) afforded **3ab** (47.4 mg, 62% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.45−7.38 (m, 4H), 7.37−7.27 (m, 7H), 7.24−7.19 (m, 1H), 7.16 (d, J = 7.9 Hz, 2H), 6.69 (s, 1H), 2.36 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.2, 137.5, 136.4, 134.1, 133.2, 129.2, 128.8, 128.5, 128.4, 127.5, 127.3, 127.0, 113.6, 97.6, 21.2.

*(1-(4-Ethylphenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3ac**).* Purification by silica gel column chromatography (PE) afforded **3ac** (33.8 mg, 57% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.46−7.37 (m, 4H), 7.36−7.23 (m, 7H), 7.22−7.13 (m, 3H), 6.67 (s, 1H), 2.64 (q, J = 7.6 Hz, 2H), 1.23 (t, J = 7.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.3, 143.9, 136.4, 134.2, 133.5, 131.8, 128.9, 128.6, 128.6, 128.5, 128.2, 127.6, 127.4, 127.1, 113.8, 97.7, 28.7, 15.70. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{23}H_{21}$, 297.1638; found, 297.1642.

*(1-(4-(tert-Butyl)phenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3ad**).* Purification by silica gel column chromatography (PE) afforded **3ad** (34,4 mg, 53% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.36−7.32 (m, 2H), 7.32−7.27 (m, 2H), 7.27−7.24 (m, 4H), 7.24−7.14 (m, 5H), 7.12−7.07 (m, J = 10.0, 8.4, 2.1 Hz, 1H), 6.60 (s, 1H), 1.24 (s, 9H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 208.4, 150.7, 136.4, 134.1, 133.2, 128.9, 128.6, 128.6, 128.2, 127.6, 127.4, 127.1, 125.5, 113.6, 97.7, 34.7, 31.4. HRMS (EI) m/z: $[M]^+$ calcd for $C_{25}H_{24}$, 324.1873; found, 324.1867.

*(1-(4-Methoxyphenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3ae** = **3ca**).*[20] Purification by silica gel column chromatography (PE) afforded **3ae** (28.6 mg, 48% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.46−7.36 (m, 4H), 7.37−7.25 (m, 7H), 7.24−7.18 (m, 1H), 6.88 (d, J = 8.8 Hz, 2H), 6.68 (s, 1H), 3.80 (s, 3H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 207.0, 158.2, 135.5, 133.1, 128.6, 127.8, 127.4, 127.4, 127.3, 126.5, 126.3, 125.9, 112.9, 112.3, 96.5, 54.3.

*(1-(4-Chlorophenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3af** = **3da**).* Purification by silica gel column chromatography (PE) afforded **3af** (32.7 mg, 54% yield) as a yellow oil. ^1H NMR (600 MHz, $CDCl_3$) δ 7.39 (d, J = 7.3 Hz, 4H), 7.37−7.28 (m, 10H), 7.25−7.22 (m, 1H), 6.71 (s, 1H).

*(1-(4-Fluorophenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3ag** = **3fa**).* Purification by silica gel column chromatography (PE) afforded **3ag** (29.8 mg, 52% yield) as a yellow oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.43−7.35 (m, 6H), 7.38−7.24 (m, 7H), 7.24−7.18 (m, 1H), 7.05−6.98 (m, 2H), 6.69 (s, 1H). $^{13}C\{^1H\}$ NMR (151 MHz, $CDCl_3$) δ 207.1, 162.1, 160.5, 135.0, 132.7, 131.1 (d, J = 3.5 Hz),

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129.0 (d, J = 8.4 Hz), 127.8, 127.6, 127.3, 126.7, 126.5, 126.0, 114.5, 114.3, 111.9, 96.8. ^{19}F NMR (376 MHz, $CDCl_3$) δ −114.49.

*(1-(4-Vinylphenyl)propa-1,2-diene-1,3-diyl)dibenzene (**3ah**).* Purification by silica gel column chromatography (PE) afforded **3ah** (21.2 mg, 36% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.46−7.36 (m, 8H), 7.36−7.27 (m, 5H), 7.24−7.20 (m, 1H), 6.71 (s, 1H), 5.75 (d, J = 17.6 Hz, 1H), 5.25 (d, J = 11.8 Hz, 1H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 208.5, 137.0, 136.4, 136.1, 135.6, 133.9, 128.9, 128.6, 128.6, 127.7, 127.4, 127.0, 126.4, 114.0, 113.6, 97.8. HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{23}H_{19}$, 295.1482; found, 295.1474.

*4-(1,3-Diphenylpropa-1,2-dien-1-yl)-1,1′-biphenyl (**3ai** = **3ga**).*[20] Purification by silica gel column chromatography (PE) afforded **3ai** (59.9 mg, 87% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.62−7.56 (m, 4H), 7.52−7.41 (m, 8H), 7.40−7.30 (m, 6H), 7.25−7.21 (m, 1H), 6.75 (s, 1H).

*2-(1,3-Diphenylpropa-1,2-dien-1-yl)naphthalene (**3aj**).* Purification by silica gel column chromatography (PE) afforded **3aj** (50.3 mg, 79% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.97−7.90 (m, 1H), 7.91−7.85 (m, 1H), 7.83−7.78 (m, 2H), 7.77−7.73 (m, 1H), 7.62−7.55 (m, 1H), 7.54−7.41 (m, 6H), 7.40−7.30 (m, 4H), 7.26−7.20 (m, 1H), 6.77 (s, 1H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 208.8, 136.3, 133.9, 133.6, 133.5, 132.9, 128.9, 128.7, 128.6, 128.2, 128.1, 127.8, 127.7, 127.5, 127.2, 127.1, 126.6, 126.3, 126.1, 125.8, 114.0, 98.0. HRMS (EI) m/z: [M]$^+$ calcd for $C_{25}H_{18}$, 318.1404; found, 318.1398.

*1-(1,3-Diphenylpropa-1,2-dien-1-yl)naphthalene (**3ak**).* Purification by silica gel column chromatography (PE) afforded **3ak** (26.3 mg, 46% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.99 (d, J = 8.1 Hz, 1H), 7.87−7.79 (m, 2H), 7.59 (d, J = 7.0 Hz, 1H), 7.52−7.33 (m, 5H), 7.32−7.11 (m, 8H), 6.67 (s, 1H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 207.5, 136.6, 134.0, 133.9, 133.8, 132.1, 128.8, 128.6, 128.4, 128.4, 127.8, 127.4, 127.3, 127.2, 127.1, 126.3, 126.2, 125.9, 125.7, 111.1, 97.4. HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{25}H_{19}$, 319.1482; found, 319.1477.

*3-(1,3-Diphenylpropa-1,2-dien-1-yl)thiophene (**3al** = **3ha**).* Purification by silica gel column chromatography (PE) afforded **3al** (25.8 mg, 47% yield) as a yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.40−7.27 (m, 9H), 7.27−7.22 (m, 1H), 7.21−7.16 (m, 1H), 6.50 (s, 1H).

*1,3-Diphenyl-1H-indene (**4**).*[19] Purification by silica gel column chromatography (PE) afforded **4** (32.7 mg, 22.0% yield) as a white solid. ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.71−7.67 (m, 2H), 7.64−7.61 (m, 1H), 7.50 (t, J = 7.5 Hz, 2H), 7.44−7.41 (m, 1H), 7.36−7.29 (m, 5H), 7.29−7.24 (m, 3H), 7.23−7.20 (m, 2H), 6.67 (d, J = 2.1 Hz, 1H), 4.74 (s, 1H).

*Benzophenone (**5**).*[16b] Purification by silica gel column chromatography (PE) afforded **5** (25.1 mg, 69% yield) as a white solid. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.82−7.79 (m, 4H), 7.62−7.56 (m, 2H), 7.49 (t, J = 7.8 Hz, 4H).

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c02708.

Experimental procedures; spectral data, including ^{1}H, ^{19}F, and ^{13}C NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

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